Intermediate Value Theorem

Exercise1: Verify that $x^3 - 2x - 5 = 0$ has a solution in the interval $[1, 2]$.

Solution:

Intermediate Value Theorem

Exercise2: Verify that $\frac{x-1}{x^2+2}=\frac{3-x}{x+1}$ has a solution in the interval [0, 3].

Solution:

Bisection Method

Exercise1: Find root of $f(x) = x^3 + 4x^2 - 10$ in $[1, 2]$ using Bisection method. Stop the calculation when $|f(c_i)| < \varepsilon$, given $\varepsilon = 0.05$. Answer in 3 decimals.

Solution:

Fixed-point Iterations

Exercise1: Find the root for $\cos x = 3x - 1$ using fixed-point iteration method with error of 0.0005

Solution:

Newton-Raphson Method

Exercise1: Find the root for $f(x) = 2x^3 + x^2 - 1$ using Newton-Raphson method with initial guess of 1 and stopping criteria of < 0.005

Solution:

Newton-Raphson Method

Exercise2: Find the root for $f(x) = \mathrm{x} - 2 + \frac{1}{lnx}$ using Newton-Raphson method with initial guess of 1.5 and stopping criteria of < 0.003

Solution: